A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 03

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
02.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (120 + 95) - (2 + 45) = 168 t ha^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO2 **1** Harvest removes biomass so less dead organic matter is returned – pasture roots and litter constantly add organic matter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Low / zero tillage **1** Adding manure, compost or mulch **1** Growing cover crops / leaving crop residues **1** Converting to long-term grass or tree crops in rotation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Spearman's rank correlation coefficient **1** Chi-squared test (frequencies / counts in categories) **1** t-test (two means of measured data) **1** Spearman's rank correlation coefficient **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Spearman** → a test for correlation between two sets of ranked data • **correlation** → a relationship in which the two variables change together • **Chi-squared** → a test comparing observed frequencies with expected frequencies **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	4
03.2	1 mark for the point below. **1** Chi-squared compares frequencies / counts in categories – it cannot be used for continuously variable measured data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **chi-squared** → a test comparing observed frequencies with expected frequencies **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	1

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 3 points below. **1** The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level **1** The negative sign shows a negative correlation: one variable decreases as the other increases **1** There is less than a 5% probability the correlation arose by chance – reject the null hypothesis (but correlation is not proof of causation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Spearman** → a test for correlation between two sets of ranked data • **critical value** → the table value the calculated statistic must reach to be significant • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	3
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Show the standard deviation / range of the original values around each mean **1** If the bars of two means overlap greatly the difference is probably not significant – little overlap suggests a significant difference **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
04.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 4.2 / 0.7 = 6 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low **1** In winter days are very short / the Sun does not rise, so light levels are extremely low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reflectors can only focus parallel rays of direct sunlight onto the absorber **1** Clouds, smoke and dust scatter the light so it cannot be focused and intensified **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	1 mark for the point below. **1** Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
04.5	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Problems:** intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature • **Photothermal:** evacuated tubes reduce conduction / convection losses – thermal stores (water, sand, concrete, molten salt) allow later use – passive solar architecture, brise soleil, light tubes – heat pumps deliver up to four times the energy used • **Photovoltaic developments:** multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces • Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency • **Integration:** grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management • **Evaluation:** strong in sunny deserts / low latitudes, weaker in high latitudes in winter – cost falling – manufacturing impacts – no single technology solves all problems – justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **pumped storage** → water pumped uphill when demand is low and released to generate at peak **Do not credit** • 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (9 marks)	
	Solar power's problems are intrinsic: it is intermittent with daily and seasonal cycles, cloud makes it unpredictable and scatters light so it cannot be focused, its low energy density demands large collector areas, the angle of incidence keeps changing, cleaning in deserts uses scarce water, manufacturing PV produces toxic wastes such as silicon tetrachloride and cadmium, and PV efficiency falls as cells heat up.	
	New technology addresses most of these. Evacuated-tube photothermal panels cut conduction and convection losses, and thermal stores of water, sand, concrete or molten salt hold heat for later; passive solar design, brise soleil and light tubes reduce demand; heat pumps deliver up to four times the energy used to run them. Multijunction PV cells absorb more wavelengths, moth-eye anti-reflective surfaces stop the 30% of light a smooth cell reflects, PV-thermal hybrids cool the cell and use the heat, transparent PV turns windows into generators, heliostats keep panels at the optimum angle and nanohydrophobic surfaces self-clean. Concentrating solar power with molten salt stored at 550 °C runs steam turbines when electricity is wanted, directly overcoming intermittency, and grid-connected farms, rooftops that avoid land conflicts, batteries and pumped storage integrate the output.	
	Even so, no technology changes the fact that winter insolation at high latitudes is tiny, that CSP needs cloudless skies, or that manufacturing has impacts. Solar can dominate supply in sunny, low-latitude regions and contribute strongly elsewhere in summer, but overcoming its problems fully depends on storage and a diverse energy mix rather than on solar technology alone.	
	Weaker answer - Level 1 (2 marks)	
	Solar power has problems because it does not work at night or when it is cloudy, and solar panels are not very efficient. New technologies can help. Better solar panels are more efficient. Solar power can be stored in batteries for use at night. Mirrors can concentrate the sunlight to make it hotter. So new technologies can overcome most of the problems with solar power.	
	What makes the difference	
	• Level 1 identifies two problems and three vague technologies; Level 3 names each technology and explains which problem it solves and how.	
	• Level 3 acknowledges what technology cannot fix (high-latitude winters, cloud for CSP, manufacturing impacts) - the 'extent' in the question.	
	• Numbers (30% reflection, 550 °C, four-fold heat-pump output) show detailed knowledge applied to the argument.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - tiny pores between clay particles allow capillary rise – none in sand	1

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1 Hygroscopic:** held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants **1 Capillary:** held in pores by surface tension – does not drain but can be absorbed by plants **1 Gravitational:** drains freely through large pores – available to plants only briefly as it passes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	1 mark for the point below. **1** Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **peds** → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Fossil fuel technologies:** secondary / tertiary recovery raising recovery from 20% to 60% – directional drilling and subsea wells – ROVs – fracking for tight oil and gas – tar sands / oil shales – coal gasification / liquefaction – enhanced gas recovery – methane hydrates – CCS (pre / post combustion, storage) • **Limits:** still finite (discoveries declining since the 1980s) – rising energy and financial cost of unconventional sources – environmental impacts (aquifers, water use, tremors, land take, habitat damage) – CO2 and other pollution – CCS unproven at scale and impractical for vehicles • **Nuclear:** new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash) – molten salt reactors – plutonium breeder and thorium reactors increasing fuel use many-fold – fusion as a long-term prospect • **Limits:** cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium – fusion not commercial • **Sustainability criteria:** maintaining supply without unacceptable environmental, economic or social impact – comparison with renewables and energy conservation – energy gap argument • **Judgement:** technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely	9

Do not write outside the box

Question	Marking guidance	Marks
	– role as transition / base-load alongside renewables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **fracking** → pumping fluid into rock at high pressure to release gas or oil • **CCS** → capturing carbon dioxide from flue gases and storing it underground • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Technology has repeatedly extended fossil-fuel supplies. Secondary and tertiary recovery raise the proportion of oil recovered from about 20% to 60%; directional drilling, subsea wells and ROVs open deep-water fields; fracking releases tight oil and gas; tar sands and oil shales, coal gasification and liquefaction convert lower-grade resources; and carbon capture and storage could reduce the CO_2 impact. Nuclear fission can be stretched even further: uranium can be adsorbed from seawater with polymers or recovered from phosphate mining and coal ash, molten-salt reactors run more safely, and plutonium breeder and thorium reactors multiply the energy obtainable from each tonne of fuel, with fusion a long-term prospect.	
	None of this makes the resources renewable. Discoveries of conventional oil have declined since the 1980s, unconventional sources cost more energy and money per unit and damage aquifers, use huge volumes of water, cause tremors and take land, and CCS is unproven at scale. Nuclear projects overrun their budgets, decommissioning and long-lived waste remain unsolved, public opposition and proliferation constrain expansion, low-grade uranium has a high embodied energy, and fusion is not commercial.	
	Sustainability means maintaining supply without unacceptable environmental, economic or social impact. New technology can extend supplies for decades and cut some impacts, which matters for closing the energy gap while renewables and conservation scale up, but it cannot make non-renewable resources sustainable indefinitely. Their proper role is as a transition and base-load source alongside renewables, not a permanent solution.	
	Weaker answer - Level 2 (5 marks)	
	New technologies mean fossil fuels can last longer. Fracking gets gas out of shale rock and directional drilling reaches oil under the sea. Carbon capture and storage can stop the carbon dioxide going into the atmosphere. For nuclear power, breeder reactors and thorium reactors can use fuel more efficiently and fusion might work in the future. However fossil fuels are still non-renewable so they will run out eventually, and fracking can pollute groundwater. Nuclear waste is still a problem. So new technologies help but they cannot make fossil fuels and nuclear power fully sustainable.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 reaches a sensible conclusion but supports it thinly; Level 3 gives specific evidence (20% to 60% recovery, declining discoveries, embodied energy of low-grade uranium).	
	• Level 3 sets out a criterion for sustainability and judges against it - the hallmark of 'evaluate'.	
	• Level 3 places the technologies in context (transition and base load alongside renewables) rather than treating them in isolation.	
06.2	1 mark for the point below. **1** 80-90% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** No reservoir range = 3.9 - 0.7 = 3.2 m – with reservoir = 2.7 - 1.7 = 1.0 m **1** Reduction = 2.2 m **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (m) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream **1** During dry periods the dam is opened to release water and maintain river levels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Deforestation / urbanisation reduce infiltration and interception **1** So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Site factors:** topography - narrow exit from a large deep basin so a small dam holds a large volume – impermeable, strong bedrock without faults or seismic activity – large catchment with regular high rainfall and low evaporation – catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation – forest / vegetation in the flooded area decaying anaerobically releasing methane – access for workers and materials and distance to demand – value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • **Alternatives for supply:** rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • **Conservation strategies:** low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • **Evaluation:** reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane – conservation is cheap and quick but limited by behaviour – transfers create conflicts between regions – desalination has high energy / carbon cost – combination needed as demand grows • Justified conclusion	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **impermeable** → does not let water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	
	Model answer - Level 3 (8 marks)	
	A reservoir site is judged first on topography: a narrow exit from a large, deep basin lets a small, cheap dam hold a large volume. The bedrock must be impermeable so water is not lost and strong enough, without faults or seismic activity, to carry the dam. The catchment should be large with regular, high rainfall and low evaporation, and its land uses should not pose a pollution risk from industry or pesticides or cause the soil erosion that would silt the reservoir. Flooding forest or vegetation causes anaerobic decay and methane. Access for workers and materials, distance to the area of demand, and the value of the land lost - urban areas and nature reserves protected, farmland less so - complete a loss-benefit analysis that different countries weigh differently.	
	Supply can also be increased without new reservoirs: rainwater collection in cities, estuary barrages, desalination where freshwater is genuinely inadequate, newly found aquifers in North Africa and Kenya, inter-basin transfers as in China and from Wales to England, artificial recharge of aquifers with wet-season surplus, river-regulation reservoirs and afforestation to even out flow. Demand can be cut by low-water appliances, xeriscaping, drip irrigation, grey-water re-use, metering and repairing the leaks that lose about a fifth of UK supply, and treatment lets contaminated sources be re-used.	
	Reservoirs give large, reliable supplies and flood control but flood land and habitats and release methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination carries a high energy and carbon cost. As demand grows, the sustainable answer is conservation first, then a combination of sources matched to each region.	
	Weaker answer - Level 2 (4 marks)	
	A reservoir needs a large catchment area with lots of rain, impermeable rock so the water does not drain away, and a narrow valley so the dam is small. It should be near the people who need the water and not flood important land. Other ways to get water include desalination, which is expensive, and collecting rainwater. Water can be conserved by fixing leaks, using water meters and having low-flush toilets. All of these help make water supplies more sustainable.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 lists correct site factors but without the reasons (why impermeable, why the valley shape matters, why land use in the catchment matters, methane from flooded vegetation).	
	• Level 3 covers far more alternatives (aquifer recharge, river regulation, transfers, afforestation) and evaluates each; Level 2 evaluates only desalination.	
	• Level 3 ends with a prioritised judgement; Level 2 ends with 'all of these help'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **stratospheric ozone** → the ozone in the stratosphere that absorbs UV	1
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms **1** Tropospheric ozone is a secondary pollutant (from NOx photochemistry) that is toxic to plants and humans and is a greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **Stratospheric ozone** → the ozone in the stratosphere that absorbs UV	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere **1** So most do not persist long enough to reach the stratosphere where UV would release their chlorine **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Screens (metal grills) trap floating / suspended non-biodegradable items (paper, plastic) which are landfilled or incinerated **1 Grit traps:** the channel widens so flow slows and stones / road grit sink out because there is not enough kinetic energy to carry them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	2
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** The effluent stands in large tanks so the faecal / organic solids sink and are removed as sludge **1** This removes over 95% of the organic matter from the fluid so the risk of deoxygenation of the river is greatly reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water	2
09.3	**1 mark each** 1 mark for each of the 3 points below. **1** Aerobic bacteria digest / break down the remaining dissolved and suspended organic matter **1** Air (or pure oxygen) is mixed in by paddle wheels or air stones so oxygen shortage does not limit the aerobic process **1** The suspended bacteria collected in the secondary sedimentation tank are the 'activated sludge' – returning them maintains a large bacterial population so digestion is rapid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.4	**1 mark each** 1 mark for each of the 3 points below. **1** Effluent from secondary treatment may still contain dissolved inorganic nutrients (phosphates, nitrates) that cause eutrophication, and some bacteria **1** Needed where the discharge site is ecologically sensitive or used by many people (bathing water, drinking water abstraction) **1** Phosphate stripping with iron(III) sulfate – micro-strainers (very fine sieves) then UV light or chlorine to kill remaining bacteria **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 269-272; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16.0 - 5.0 = £11 million **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (£ million) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
10.2	**1 mark each** 1 mark for each of the 3 points below. **1** It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided **1 ALARA:** emissions should be As Low As Reasonably Achievable **1 BATNEEC:** the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** As industries expand or cities grow the number of pollutant sources increases **1** So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Critical Pathway Analysis:** predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk – mainly radioactive discharges – checked by sampling • **Critical Group Monitoring:** exposure of the most-exposed group monitored before health effects – if their risk is acceptable everyone else is safer – emissions reduced if not • **Emission control strategies:** location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) • **Principles:** polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship • **Selection of methods:** production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) • **Evaluation:** prevention usually cheaper and more effective than remediation – costs rise steeply near 100% – CPA relies on understanding of pathways – enforcement and international agreement needed for transboundary pollutants – judgement about acceptable risk is political as well as scientific • Justified conclusion	9

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Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **persistent** → does not break down / stays in the environment for a long time • **bioaccumulation** → a pollutant building up inside one organism over time • **Critical Group** → the people most at risk / most heavily exposed **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Deciding how much to control pollution starts with predicting where a pollutant will go. Critical Pathway Analysis uses the state of the effluent, wind and currents, geology, the effect of pH and oxygen on solubility, bioaccumulation and local food sources to predict whether a discharge will disperse harmlessly or concentrate somewhere sensitive; it is mainly used for radioactive discharges and means monitoring can focus on the sites at risk rather than sampling everywhere. Critical Group Monitoring then checks the exposure of the people most at risk from their lifestyle before any health effects occur: if their risk is acceptable everyone else's is lower, and if not, emissions are reduced.	
	The choice of control is guided by principles. The polluter pays principle gives a financial incentive to prevent damage, but only if penalties are high and polluters cannot escape responsibility. The precautionary principle assumes a waste is harmful until research shows otherwise, so ignorance is no defence. Because the cost of control rises steeply as emissions approach zero, emissions are set As Low As Reasonably Achievable using the Best Available Technology Not Entailing Excessive Cost. Where emission location and timing can be chosen (marine outfalls, downwind of towns, not over aquifers; avoiding temperature inversions) severity is reduced cheaply.	
	Overall, prevention of production (fuel desulfurisation) or of release (electrostatic precipitators, catalytic converters) is more effective than post-release remediation such as oil-spill clean-up, which never recovers all the pollutant. CPA depends on understanding pathways, and transboundary pollutants need international enforcement, but together these principles give a rational basis for deciding acceptable risk rather than aiming for an unaffordable zero.	
	Weaker answer - Level 1 (3 marks)	
	Pollution can be controlled in lots of ways. Critical Pathway Analysis looks at where pollution goes and Critical Group Monitoring looks at people. The polluter pays principle means the person who pollutes has to pay. The precautionary principle means being careful. Pollution can be prevented, or cleaned up afterwards with things like booms for oil spills. Electric cars are an alternative process. Pollution control can be expensive so ALARA is used. All of these are good ways of controlling pollution.	
	What makes the difference	

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Question	Marking guidance	Marks
	• The Level 3 answer explains *how* each principle works (why penalties matter, why costs rise near zero emissions); the weak answer only names them.	
	• The Level 3 answer groups the material logically (predict, decide, select) rather than listing everything at the same weight.	
	• 'Evaluate' needs comparison and a judgement: prevention vs remediation, CPA's dependence on understanding pathways, and a conclusion about acceptable risk. The weak answer ends with 'all of these are good'.	
	• Precise examples (desulfurisation, precipitators, catalytic converters, temperature inversions) show applied knowledge; 'things like booms' does not.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

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Question	Marking guidance	Marks
11.1	**Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Negative feedback:** increased low-level cloud (albedo), increased photosynthesis and sequestration – hydrological cycle evaporation-precipitation • **Positive feedback:** melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • **Tipping points:** self-sustaining processes – why control must precede them – examples • **Implications:** urgency and precautionary principle – need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation – geoengineering risks • Difficulty of prediction because feedbacks interact at different scales – models – international agreements and per capita footprints • Judgement on whether current strategies are adequate	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	

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Question	Marking guidance	Marks
11.2	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

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